NATIONAL INSTITUTE OF STATISTICS OF RWANDA (NISR) BIG DATA HACKATHON 2026

Rwanda Yield Optimizer: Data-Driven Yield Gap & Climate Resilience Platform

Executive Concept Note & Policy Alignment Framework Aligning Data Fusion & Machine Learning with NST2 & PSTA 5

EXECUTIVE SUMMARY

Rwanda Yield Optimizer Platform is an end-to-end decision-support system designed to accelerate Rwanda's agricultural productivity and climate adaptation targets under PSTA 5 and NST2. By fusing official NISR baseline survey microdata (SAS 2025, AHS 2024, EICV7) with real-time Open-Meteo climate feeds, Sentinel-2 satellite vegetation indices (NDVI), and eSoko commodity prices across 6 focal districts, the system powers a Champion Gradient Boosting Regressor ($R^2 = 0.8116$). The platform equips policy officers at MINAGRI, RAB, and district offices with an interactive web dashboard featuring real-time yield scenario optimization, climate vulnerability rankings, and actionable resource allocation frameworks.

1. Problem Statement & Socio-Economic Context

Agriculture engages 3.6 million adults across 2.2 million households (65.3% nationwide) according to the NISR AHS 2024 report. Despite serving as the bedrock of food security and rural livelihoods, smallholder productivity is acutely constrained by four structural bottlenecks:

- Substantial District Yield Gaps: Official SAS 2025 data reveals severe geographic yield disparities. While high-altitude zones like Musanze (58,546 MT Season A Irish Potato) achieve high productivity, drier arable districts face severe yield deficits.
- Severe Water Deficits & Low Irrigation Reach: AHS 2024 highlights that only 14.1% of agricultural households practice irrigation (falling to 7.8% in Musanze and 3.9% in Burera). Open-Meteo climate data records an extreme annual water deficit of 916.8 mm in Gatsibo, creating acute seasonal moisture stress.
- Land Constraints & Tenure Pressures: 71.8% of farming households cultivate plots under 0.5 hectares, and 50.6% operate on rented land, disincentivizing long-term soil conservation and infrastructure investments.
- Socio-Economic Disparities: EICV7 Poverty Profile data indicates a national poverty rate of 27.4%, with district rates spanning from 11.1% in Gasabo to 24.2% in Huye and 44.2% in Rusizi, underscoring the need for geographically targeted interventions.

2. Strategic Alignment with PSTA 5 & NST2

The platform converts raw statistical microdata into proactive policy intelligence directly aligned with national strategic roadmaps:

- PSTA 5 (Strategic Plan for Agriculture Transformation): Directly advances Pillar 1 (Production & Productivity) by quantifying yield gaps, optimizing inorganic fertilizer and Smart Nkunganire subsidy reach (currently 16.6% nationwide), and targeting post-harvest loss mitigation.
- NST2 (Second National Strategy for Transformation): Supports sustainable modernization by expanding climate-smart irrigation infrastructure, strengthening extension service reach (currently 67.1%), and enhancing rural household resilience.
- Vision 2050: Establishes evidence-based decision tools to transition subsistence farming into commercialized, high-value agriculture.

3. Integrated Multi-Source Data Architecture

The platform synthesizes four complementary data streams into a unified feature matrix across 6 target focal districts (Burera, Gatsibo, Musanze, Huye, Rusizi, Gasabo):

Data Stream	Primary Source	Role in Analytics & Modeling Engine
Baseline Crop Production	SAS 2025 Report	Ground truth target metrics for yield modeling
Household & Inputs	AHS 2024 Report	Socio-economic readiness & input efficiency features
Poverty & Welfare	EICV7 Survey	Social vulnerability weighting & target prioritization
Farmgate Prices	eSoko (MINAGRI)	Economic revenue simulation & post-harvest

		valuation
Agro-Climate API	Open-Meteo API	Environmental constraint & moisture shortage feature
Geospatial Satellite	Sentinel-2 (GEE)	Real-time crop vigor & canopy health monitoring

4. Machine Learning Yield Modeling & Empirical Performance

To ensure high predictive accuracy without data leakage on the integrated 36 district-crop observation matrix (6 focal districts across 6 crops: Irish Potato, Maize, Beans, Cassava, Rice, Amaranth), all models were rigorously evaluated using Leave-One-Out Cross-Validation (LOOCV). Five distinct ML architectures were benchmarked:

Model Architecture	R^2 Score	RMSE (kg/ha)	Validation Status & Selection Rationale
Gradient Boosting Regressor	0.8116	1,920.43	Champion Model: Lowest error, robust non-linear capture
XGBoost Regressor (Tuned)	0.7965	1,995.43	High Performer: Strong gradient tree ensemble baseline
Ridge Regression (Tuned)	0.7307	2,295.84	Regularized Linear Baseline: Multi-collinearity controlled
Random Forest	0.7088	2,387.42	Ensemble Baseline:

Regressor			Captures general feature splits
Linear Regression (OLS)	0.6900	2,463.24	Unregularized Linear Baseline

Key Empirical Insights & Feature Importance Rankings:

- Water Deficit (mm) & Moisture Shortage Index: Extracted as the #1 predictive constraint on crop yields nationwide. Districts with severe moisture deficits (e.g., Gatsibo at 916.8 mm) experience significant yield penalties unless buffered by irrigation.
- Sentinel-2 Satellite NDVI: Canopy vegetation greenness in Season B directly correlates with final biomass accumulation.
- Extension Service Reach (%): Institutional advisory contact significantly cushions yields against environmental stress factors.
- Yield Prediction Precision: On high-yielding staple crops like Cassava (~14,000 kg/ha) and Irish Potato (~12,000 kg/ha), the Champion MAE of 1,222.41 kg/ha represents an exceptionally low relative error margin (~8–10%).

5. Live Interactive Web Dashboard & Decision Support

The platform is deployed live on Streamlit Community Cloud and organized into 4 functional decision-support modules:

- Tab 1: District Productivity & Yield Gap Explorer: Interactive multi-filters mapping baseline production (MT), yields (kg/ha), and post-harvest loss rates across all focal districts.
- Tab 2: What-If Yield Optimizer & Scenario Simulator: Interactive policy sliders (Irrigation Expansion +0-50%, Extension Reach +0-40%, Fertilizer Adoption +0-50%) feeding real-time inputs into the Champion ML model to calculate projected yield increases (MT) and economic revenue gains in Rwandan Francs (RWF).

- Tab 3: Climate Vulnerability & Irrigation Priority Index: Visual ranking combining water deficit, poverty headcount %, and post-harvest loss to pinpoint priority infrastructure zones.
- Tab 4: Policy Support & Data Export: Automated NST2 policy guidance and downloadable CSV datasets.

6. Actionable Policy Recommendations & National Impact

Based on empirical model findings and district vulnerability rankings, four priority policy interventions are established:

- Prioritize Small-Scale Water Harvesting in Gatsibo: Deploy targeted irrigation dams and water catchment infrastructure to offset Gatsibo's 916.8 mm moisture deficit.
- Expand Extension Advisory in High-Yield Hubs: Scale Smart Nkunganire subsidy distribution and extension reach in Musanze and Burera to maximize high-altitude potato yields.
- Post-Harvest Loss Mitigation in Rice & Maize: Deploy modern drying and storage facilities in Rusizi and Huye to preserve harvested volume and reduce post-harvest waste.
- Scalable Digital Public Good: The entire codebase and pipeline are open-source, fully reproducible, and ready for national scale-out across all 30 districts.